Evaluating XAI Support From A Hierarchical Reinforcement Learning Policy in Human-Agent Collaboration

Full version of paper accepted as an Extended Abstract at AAMAS 2026

Mateus Levi Simões Fernandes
PUC-Rio
Rio de Janeiro, Brazil
mfernandes@inf.puc-rio.br

Alberto Sardinha
PUC-Rio
Rio de Janeiro, Brazil
sardinha@inf.puc-rio.br

ABSTRACT

Explainable AI (XAI) has shown promise for human-agent collaboration, yet results rely on evaluations of hand-crafted policies in custom environments, limiting generalizability to state-of-the-art teaming research. We provide the first systematic evaluation of XAI support generated from an intrinsically explainable machine learning policy in an established human-agent teaming benchmark. Using the Hierarchical Ad Hoc Agents (HA^2) architecture in Overcooked-AI, we generate real-time explanations from hierarchical subtask selections, delivered through text or audio modalities via a novel trigger-based system. Our between-subjects experiment (n=38) found no statistically significant performance benefits from these explanations, challenging assumptions about XAI utility in fast-paced collaboration. However, audio explanations show trends toward faster learning rates, suggesting explanations may enhance adaptation rather than immediate performance. We provide the first modality comparison in real-time human-agent collaboration and establish a baseline methodology for evaluating intrinsically explainable reinforcement learning architectures in benchmark environments. With post-hoc analysis revealing high gaming familiarity in our sample, our findings suggest that XAI effectiveness depends critically on matching explanation sophistication to user expertise and task demands, bridging XAI and human-agent teaming research.

CCS CONCEPTS

• **Human-centered computing** → *Auditory feedback*; **Empirical studies in HCI**; Natural language interfaces; *User studies*; • **Computing methodologies** → **Cooperation and coordination**; • **General and reference** → **Empirical studies**; • **Theory of computation** → **Reinforcement learning**.

KEYWORDS

Human-Agent Teaming, Hierarchical Reinforcement Learning, XAI

1 INTRODUCTION

As artificial intelligence systems transition from tools to collaborative partners, they must become not only effective in executing tasks but also comprehensible to their human teammates. In human-human collaboration, teammates communicate intentions and negotiate strategies using shared mental models. But human-AI collaboration often lacks mutual understanding due to agents having opaque decision-making processes, frequently based on hard-to-comprehend neural network architectures. This understanding gap contradicts McDermott's observation that systems must be both correct *and* understood [8], creating a fundamental challenge: if AI agents are to occupy roles traditionally reserved for humans, they must operate at levels of transparency similar to human teammates.

Explainable artificial intelligence (XAI) research offers promising approaches to bridge this gap [11], yet systematic evaluation of such techniques in collaborative contexts remains limited. Most human-agent teaming research prioritizes optimizing agent performance over leveraging human understanding [7, 13], treating humans as unpredictable variables rather than partners whose comprehension of the autonomous agent could enhance coordination. While Paleja et al. [10] demonstrated that XAI-based support can improve collaboration, their work relied on hand-crafted policies in a custom Minecraft environment. This limits transferability of their conclusions to state-of-the-art human-agent teaming research, which uses established benchmarks like Overcooked-AI [3]. Such a context reveals a critical methodological gap: no work has systematically evaluated XAI support generated from learned, high-performing policies in benchmark collaborative environments.

We address this gap by leveraging the Hierarchical Ad Hoc Agents (HA^2) [1] architecture, which achieves state-of-the-art performance in Overcooked-AI through policies trained on a human-interpretable subtask decomposition. HA^2's hierarchical structure displays intrinsic interpretability: any action taken by the agent on the environment depends on comprehensible high-level subtask choices. Even so, this explanatory potential has not been systematically evaluated by the authors or succeeding work. To remedy this, we generate real-time explanations directly from the agent's hierarchical decision-making process and deliver them through text or audio modalities via a novel trigger-based system.

Our between-subjects experiment (n=38) examines both immediate effects and adaptation patterns across four collaborative sessions, investigating whether explanations improve objective performance and subjective metrics, and whether audio delivery reduces cognitive interference in visually demanding tasks. Our findings reveal that simple status explanations provide no performance benefits for participants, yet audio explanations show trends toward faster learning rates, suggesting XAI may enhance adaptation rather than immediate performance. Given the exploratory nature of this first systematic evaluation of intrinsically explainable reinforcement learning policies in an established HAT benchmark, we emphasize effect size estimation alongside significance testing, with findings positioned as baseline patterns to guide future adequately-powered studies. Our sample's high mean gaming familiarity (revealed through post-hoc analysis) provides important context for interpreting these exploratory results.

This work makes three key contributions: (1) the first systematic XAI evaluation using state-of-the-art learned policies in an established teaming benchmark, demonstrating that intrinsically explainable reinforcement learning architectures can generate authentic explanations for real-time collaboration; (2) the first comparison of explanation modality effects (text vs. audio) in a real-time human-agent collaborative task; and (3) a baseline trigger-based delivery system for hierarchical policy explanations. These insights advance understanding of when and how XAI operates in fast-paced collaborative environments, providing guidance for designing explanation systems that account for user expertise and task demands as well as for future applications of XAI support for human-agent teaming.

2 RELATED WORK

The challenge of making AI agents comprehensible during collaboration sits at the intersection of two research communities with complementary blindspots. Explainable AI research has developed sophisticated techniques for transparency but struggles with systematic evaluation in realistic collaborative contexts. Human-agent teaming research has created benchmark environments and state-of-the-art learning algorithms but largely ignores the social dimensions of collaboration, treating agent transparency as secondary to performance optimization.

XAI in Human-Agent Collaboration: One of the most comprehensive evaluations of XAI utility during real-time human-agent collaboration is found in Paleja et al.'s using Minecraft [10], which demonstrated that the effectiveness of XAI support depends critically on user expertise and the type of explanation given by the agent. Novice participants showed improved performance when receiving status-based explanations, but not with full decision tree visualizations. Meanwhile, expert participants experienced performance degradation when receiving explanations, presumably due to higher cognitive load disrupting fluid coordination. Subjective measures told a different story: participants consistently rated explainable agents more positively regardless of performance effects, suggesting explanations can still provide subjective benefits beyond immediate task optimization.

Paleja's work established important principles: that real-time explanations differ fundamentally from post-hoc analysis; that modality and form of the explanations matter for fast-paced tasks; and that task expertise moderates explanation utility. Yet two critical limitations constrain its applicability to modern human-agent teaming research. First, the agent used a hand-crafted hierarchical policy rather than learned behaviors, making it unclear whether their findings transfer to deployable machine learning systems. Hand-crafted policies enable perfect control over explanation content and timing but sidestep the fundamental challenges of generating explanations from learned decision-making processes. Second, the work was based on a custom Minecraft environment where the agent had limited capabilities – serving rather as an assistant gathering or constructing resources for the human participant – instead of being an equal partner. This limits generalizability when we consider established collaborative benchmarks, where coordination faces additional challenges and (or because) teammates have equal capabilities.

Recent work by Wang et al. [15] addresses some of these limitations by developing external Theory of Mind models that predict future actions of black-box agents in the benchmark Overcooked-AI environment. By training transformer-based predictors on offline agent trajectories, their approach generates visual predictions of upcoming primitive actions without requiring the agent itself to be interpretable. While this demonstrates that post-hoc explanation systems can improve human-AI coordination even for opaque agents, their approach relies on substantial offline data collection and separate model training pipelines – factors that are avoidable if a model's intrinsic explainability can be leveraged. The visual prediction modality they employ also differs from simpler explanation approaches based on natural language, leaving open questions about how different modality and content choices would affect explanation utility.

Prior work exploring communication in collaborative contexts reinforces both the promise and challenges of agent transparency. Early work by Harbers et al. [4] found that agent explanations improved subjective experience without enhancing performance, attributing this to cognitive overhead. Mercado et al. [9] showed that transparent AI assistants increased both trust and performance in unmanned vehicle operations, though using domain-specific transparency models. More recently, Zhang et al. [16] demonstrated through a Wizard-of-Oz study that humans prefer proactive AI communication while noting that excessive communication creates distraction, particularly after coordination patterns stabilize. These findings collectively suggest that explanation utility depends on matching communication frequency and content to both task demands and human expertise levels, but none of these works evaluated explanations generated from state-of-the-art learned policies in a benchmark teaming environment.

Human-Agent Collaboration in Overcooked-AI: Carroll et al. [3] established Overcooked-AI as the *de facto* benchmark for human-agent collaboration research, demonstrating that agents trained exclusively with other AI partners fail catastrophically when teaming with humans. Their Behavior-Cloning Play (BCP) method addressed this by training agents on human gameplay data, but subsequent work challenged the necessity of human data entirely. Strouse et al.'s Fictitious Co-Play (FCP) [13] achieved superior performance by training agents as best responses to diverse self-play populations rather than specific human behaviors, establishing that exposure to varied teammate profiles enables effective human collaboration without the necessity of data generated from human behavior.

This insight that robustness to diverse teammates enables human collaboration motivated increasingly sophisticated population-based training approaches. Loo et al.'s Hierarchical Population Training (HiPT) [7] extended FCP by learning multiple low-level policies alongside a high-level manager that dynamically selects the appropriate policy which serves as the best response for the current teammate. This meta-learning approach, which trains the agent to recognize and adapt to different teammate types, achieved state-of-the-art performance by making the agent robust not just to diverse behaviors but to varying skill levels.

The Hierarchical Ad Hoc Agents (HA^2) architecture [1] represents a conceptual shift from pure performance optimization toward human-aligned task representations. Rather than learning multiple best response policies like HiPT, HA^2 provides agents with the same hierarchical task structures humans naturally use for coordination, decomposing complex collaborative tasks into mutually understandable subtasks and using this hierarchy as basis for the agent's decision-making. This approach leverages the insight that effective human collaboration emerges not just from adapting to diverse behaviors but from operating within shared cognitive frameworks [12, 14].

Critically for our purposes, HA^2 achieves this human alignment while maintaining state-of-the-art performance in Overcooked-AI through a Manager-Worker architecture. The Manager selects high-level subtasks (e.g., "pick up onion from dispenser", "place onion in pot") while a Worker policy executes the low-level actions to complete the chosen subtask. The authors argue this hierarchical structure inherently supports transparency: any action the agent takes can be traced to a human-interpretable subtask choice. Yet this claim of interpretability has never been empirically validated, since the original work focused exclusively on implicit coordination through shared task abstractions without evaluating whether explicitly communicating these subtask choices actually improves human-agent collaboration.

This creates our core opportunity: HA^2 provides both high performance and intrinsic explainability through its hierarchical decomposition. Unlike hand-crafted policies [10] that sacrifice deployability for interpretability or opaque deep learning policies that require external models and separate training [15], HA^2's architecture naturally generates human-interpretable decision representations as part of its normal operation. The Manager's subtask selections occur at a level of temporal and semantic abstraction that matches how humans naturally describe collaborative intentions: "I'm getting an onion" rather than "moving south three tiles then pressing interact."

By exposing these subtask choices through real-time explanations, we can evaluate XAI utility using a state-of-the-art learned policy in an established benchmark environment. Furthermore, we systematically compare text versus audio delivery to understand how modality choices affect explanation utility in visually demanding tasks – something prior work has not evaluated despite using different modalities [10, 15]. This methodological advance enables more rigorous investigation of when and how XAI operates in realistic human-agent teaming contexts.

This review reveals three unexplored opportunities our work addresses. First, no systematic XAI evaluation exists in established HAT benchmarks. While Paleja et al. demonstrated promising results, their custom environment and hand-crafted policy limit comparison with state-of-the-art approaches. Overcooked-AI provides the benchmark context necessary for systematic evaluation, with established baselines and metrics enabling direct comparison of XAI approaches. Second, intrinsically explainable learned policies remain unexplored for real-time collaboration. Previous work used either Wizard-of-Oz methods without actual ML policies [16], hand-crafted policies that don't effectively represent deployable systems [10] or post-hoc explanation systems applied on opaque policies [15]. HA^2's architecture bridges this gap, since its hierarchical structure enables explanation generation while its learned policies achieve competitive performance. Third, explanation modality effects in fast-paced environments remain largely unexplored. While some explanation systems have been evaluated separately on different collaborative environments, systematic comparisons between explanation modalities (without content differences) in benchmark environments remain absent from the literature. Real-time collaboration may benefit, for example, from audio delivery – which avoids competing for visual attention during dynamic tasks, yet this has not been empirically evaluated.

By generating explanations directly from HA^2's subtask selections and comparing text versus audio modalities across multiple sessions, we provide the first systematic evaluation of XAI-based support from state-of-the-art learned policies in an established collaborative benchmark.

3 METHODOLOGY

This work investigates two primary research questions that address critical gaps in explainable AI for human-agent collaboration. First, we examine how XAI-based support generated from intrinsic explainability impacts human-agent collaboration in fast-paced environments (**RQ1**), where the temporal demands of real-time coordination may fundamentally alter explanation utility compared to slower-paced tasks. Second, we investigate whether explanation modality (specifically text versus audio delivery) affects collaboration outcomes (**RQ2**), recognizing that modality choice may be particularly consequential in visually demanding tasks where text explanations compete for limited attentional resources.

Drawing on prior literature, we formulated three hypotheses to guide our investigation. We hypothesize that explanations will improve team performance compared to no explanations (**H1**), based on evidence that transparency facilitates coordination in collaborative contexts. We further hypothesize that audio explanations will reduce cognitive overhead compared to text in visually demanding tasks (**H2**), as audio delivery may allow participants to process agent intentions without shifting visual attention from the primary task. Finally, we hypothesize that explanations will improve subjective agent perception (**H3**), consistent with findings that explainable agents are rated more positively regardless of performance effects.

We generate real-time explanations from the Hierarchical Ad Hoc Agents (HA^2) architecture and evaluate their impact on human-agent collaboration in Overcooked-AI across text and audio modalities. This section establishes the theoretical foundations for explanation generation, describes our agent implementation, and details the experimental design.

3.1 Theoretical Foundations

Collaborative tasks in Overcooked-AI are formalized as Decentralized Partially Observable Markov Decision Processes (Dec-POMDPs), defined by the tuple $\{S, A, P, r, O, \Omega, N, \gamma\}$ where S is the shared state space, $A = A_1 \times A_2 \times \cdots \times A_N$ is the joint action space, $P(s'|s, a)$ defines state transitions, $r(s, a)$ provides shared rewards (appropriate for fully cooperative settings), O represents agent observation spaces, $\Omega(o|s', a)$ gives observation probabilities, N is the number of agents, and $\gamma \in [0, 1]$ is the discount factor. Partial observability arises primarily from decentralized decision-making: while agents observe similar environmental states, each must commit to actions without knowing their teammate's concurrent choice or internal decision-making process.

To enable explanation generation, we leverage hierarchical reinforcement learning policies that decompose agent behavior into interpretable levels. Our agent uses:

- **Subtask Space** $Z = \{z_1, z_2, \ldots, z_K\}$: A predefined set of semantically meaningful goal states that correspond to human-interpretable collaborative objectives (e.g., "pick up onion from dispenser", "place onion in pot").
- **High-Level Policy** $\pi_H(z|o)$: O $\rightarrow$ Z: Selects appropriate subtasks given current observations, with subtask selections reflecting strategic decisions about task priorities and coordination.
- **Low-Level Policies** $\pi_W^i(a|o,z)$: O $\times$ Z $\rightarrow$ A: Execute primitive environment actions to complete assigned subtasks, translating high-level intentions into concrete behaviors.

This hierarchical decomposition enables natural explanation generation because subtask selections operate at a level of abstraction that aligns with human task understanding. Unlike primitive actions, which represent moment-to-moment decisions, subtasks capture meaningful intentions that may persist over multiple timesteps. When the Manager selects subtask z_i, this decision can be translated into human-understandable explanations ("collecting ingredients", "serving soup") rather than streams of primitive action descriptions.

However, a critical constraint must be acknowledged: when hierarchical policies use model-free reinforcement learning, the agent lacks explicit world models or planning capabilities. The Manager policy selects subtasks *reactively* based on current observations, not through deliberative planning. Consequently, subtask selections represent the agent's current high-level intention at the moment of explanation, not commitments to future behavior. The agent may change its subtask selection in subsequent timesteps as new observations arrive, without internal representations of why changes occurred or what alternatives were considered. Despite this limitation, hierarchical task decomposition allows us to provide explanations that reflect the agent's actual decision-making process while still making sense to human teammates without the necessity of post-hoc rationalizations.

3.2 Agent Implementation

The HA^2 framework [1] implements the hierarchical structure described above through a Manager-Worker architecture. The Manager policy $\pi_H(z|o)$ selects from twelve domain-specific subtasks in Overcooked-AI: (1) pick up onion from dispenser, (2) pick up onion from counter, (3) pick up dish from dispenser, (4) pick up dish from counter, (5) place onion in pot, (6) place onion on counter, (7) get soup from pot, (8) place dish on counter, (9) get soup from counter, (10) place soup on counter, (11) serve soup, and (12) waiting/unknown.

We use the original HA^2 Worker policies provided by the authors as pre-trained PyTorch models. For the Manager, we trained a custom policy using Proximal Policy Optimization (PPO) and self-play specifically on the Counter-Circuit layout. During training, both agents received their environmental observation plus their teammate's current subtask choice, enabling the Manager to potentially adapt its decisions based on the teammate's high-level intention. The Manager policy runs every timestep, outputting subtask choices that define the Worker policy responsible for primitive action selection at each timestep.

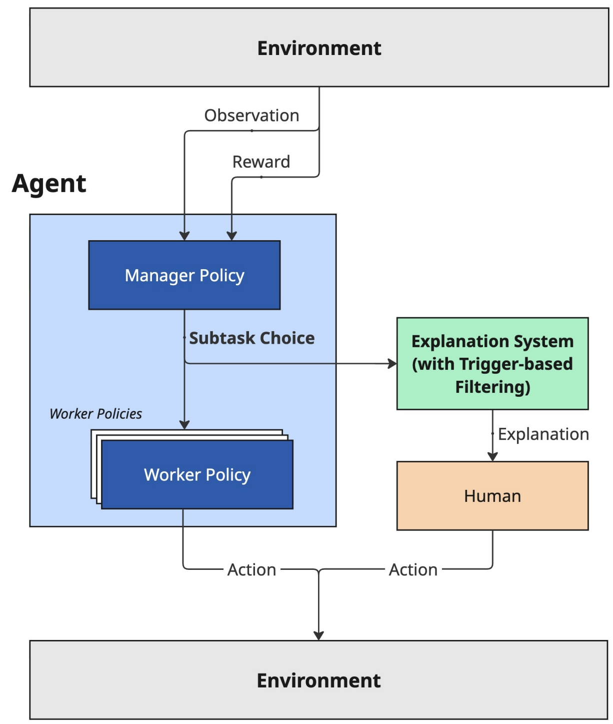

Figure 1: System architecture demonstrating how the Manager's subtask selections serve dual purposes in our XAI approach: directing the Worker's primitive actions and generating human-interpretable explanations. The trigger-based filtering system ensures explanations are delivered only at coordination-relevant moments rather than every timestep.

3.3 Explanation Delivery System

Converting Manager subtask selections into human-directed explanations required addressing temporal dynamics revealed through pilot testing with eight graduate students. As illustrated in Figure 1, the Manager's subtask outputs flow both to the Worker policy (for action execution) and to our explanation generation system (for human communication). Pilot testing revealed critical issues with continuous explanation display: first, explanations generated every timestep made it difficult for participants to understand them, let alone react accordingly. Second, minor policy fluctuations generated explanations with no coordination value, and participants reported focusing on gameplay rather than processing the constant stream of information. Continuously displaying Manager outputs also proved overwhelming, since the high-level policy runs every timestep and frequently changes selections even during stable task execution. While this behavior is natural for model-free policies, it proved to be confusing when communicated as human-like intentions. These findings motivated our trigger-based approach that generates explanations only at coordination-relevant moments.

We implemented a priority-based trigger system that generates explanations primarily in moments where revealing the agent's high-level intention is relevant to its teammate:

(1) **Blocking**: Triggers when the human occupies the agent's target tile or stands one tile away from its intended path, communicating navigation intent through templates like "Moving around you to [subtask]" or "Excuse me, I want to [subtask]."
(2) **Critical Path**: Triggers when the agent selects subtasks essential for task progression (e.g., placing the final onion needed to complete a soup or collecting a ready dish from a pot), using templates such as "I'll [subtask]" to signal strategic coordination points.
(3) **Distance Threshold**: Triggers when the agent travels a predetermined number of tiles since the last explanation, providing status updates like "I'm going to [subtask]" during extended movement sequences.
(4) **Subtask Change**: Triggers when the Manager switches to a new subtask (excluding transient "unknown" states during policy transitions), announcing behavioral shifts with templates like "Now I'll [subtask]."

Each trigger maps to natural language templates that frame subtask descriptions appropriately for their coordination context (see Appendix A of supplementary materials for complete template specifications). A six-second cooldown prevents excessive communication regardless of trigger activation frequency. In the case of the text condition, each explanation remains visible for four seconds.

Modality Implementation: We deliver identical explanation content through two modalities. The **Text condition** displays explanations in a semi-transparent black bar at the bottom of the game interface, prefixed with "Agent:" in white text. The **Audio condition** converts identical text to speech using browser-native synthesis with automatic voice selection based on participant language preference (Portuguese or English) and speech rate optimized during pilot testing. Audio condition participants completed a validation task before gameplay: correctly entering a randomly generated number spoken by the text-to-speech system confirmed audio functionality. Text condition participants viewed a sample interface screenshot showing explanation placement.

3.4 Experimental Design

Our between-subjects design assigned participants to three conditions via blocked randomization (block size 6, two participants per condition per block): **Control** (no explanations), **Text**, and **Audio**. Each participant completed four 80-second gameplay sessions in Overcooked-AI's Counter-Circuit layout at 10 FPS. We selected Counter-Circuit for three reasons: (1) coordination complexity – the central counter obstacle requires continuous movement coordination and creates natural blocking scenarios where explanations could aid navigation; (2) strategic depth – optimal performance requires dividing labor around the counter, enabling assessment of whether explanations facilitate role negotiation; (3) experimental control – restricting to one layout eliminates layout-specific effects

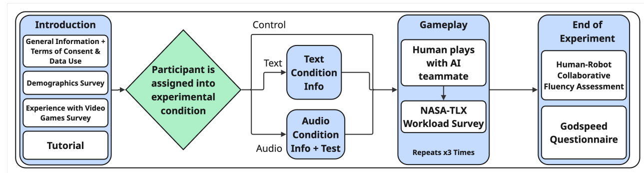

Figure 2: Experimental procedure flow showing blocked randomization for condition assignment. Participants complete introduction surveys before being randomly assigned to Control, Text, or Audio conditions, then play four gameplay sessions with intermediate assessments, followed by post-experiment questionnaires.

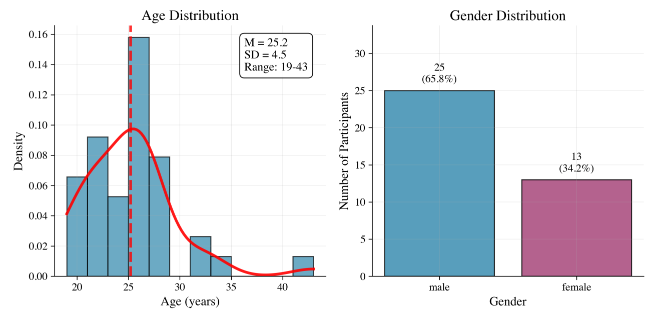

Figure 3: Participant demographics showing a young adult sample (M=25.2 years, SD=4.5) with male majority (≈66%). Age distribution spans 19-43 years with peak density around 25, while gender distribution shows 25 male and 13 female participants.

as a confounding variable for this initial study on the impact of XAI support.

After each session, participants completed the NASA-TLX workload assessment [5]. Following all gameplay, they evaluated the agent using the Godspeed Questionnaire [2] (measuring anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety) and a Human-Agent Fluency Assessment [6] adapted from human-robot teaming research by replacing "robot" with "agent." Instructional manipulation checks embedded in the first and third NASA-TLX surveys and both post-experiment questionnaires verified participant attention. Complete survey instruments are provided in Appendix B of the supplementary materials.

We developed a web-based platform integrating Overcooked-AI with our explanation delivery system. The platform consists of a Flask backend with Firebase Firestore for data persistence and a browser-based frontend rendering gameplay via Phaser.js. The system was deployed on a virtual machine, supporting up to 15 concurrent sessions for remote participation. We collected game scores, participant action counts, network latency metrics, and complete survey responses. Participants achieving less than 80% accuracy on instructional manipulation checks or experiencing technical issues (incomplete sessions, extreme network latency) were excluded from analysis.

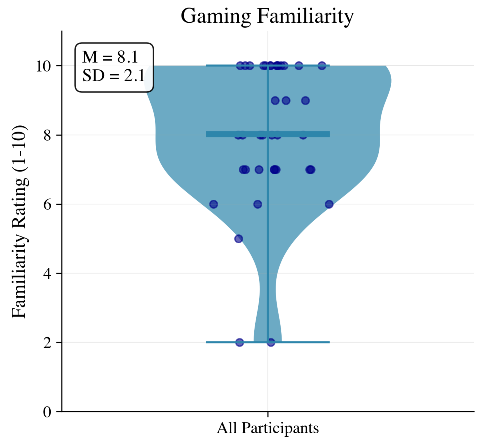

Figure 4: Gaming familiarity distribution demonstrating a gaming-expert sample. Participants reported high familiarity scores (M=8.1 on 1-10 scale, SD=2.1), with most clustered between 7-10, indicating substantial videogame experience that may influence explanation utility.

3.5 Participants and Procedure

For our IRB-approved experimental study, we targeted a sample size of n=30 based on similar studies in human-agent collaboration research [10] and practical recruitment constraints, expecting to detect large effects given the apparentness of the interventions. Anticipating potential connectivity issues during remote participation, we recruited 41 participants via convenience sampling within the local academic community (departmental emails and social media groups), yielding 38 valid participants after quality screening based on minimum accuracy on survey attention checks.

Post-hoc sensitivity analysis revealed our final sample provided 80% power to detect large effects for H1 (Cohen's d≥0.86), very large effects for H2 (d≥1.12), and large effects for H3 (Cohen's f≥0.53). Our observed effects were smaller – H1: d=0.51 (medium), H2: d=0.14 (negligible) – confirming we were underpowered for medium effects. We therefore frame this work as exploratory research establishing baseline effect sizes for future confirmatory studies.

Participants accessed a web-based platform, completing: (1) consent and demographics surveys, (2) videogame experience assessment, (3) tutorial video and text instructions, (4) condition assignment and modality-specific instructions, (5) four gameplay sessions with intermediate NASA-TLX assessments, and (6) post-experiment questionnaires. The complete experimental flow is shown in Figure 2.

Our final sample comprised 38 participants (ages 19-43, M=25.2, SD=4.5; approximately 66% male; Control n=14, Text n=14, Audio n=10), as shown in Figure 3. Participants reported high gaming familiarity (M=8.1 on a 1-10 scale, SD=2.1), visualized in Figure 4, with 92% self-reporting a score ≥ 6/10 and 84% playing video games at least weekly. Unlike Paleja et al. [10], who used post-hoc performance clustering, we characterize participants based on this self-reported gaming familiarity, revealing a predominantly gaming-experienced sample. No significant baseline differences emerged across conditions in age, gaming familiarity, or gender distribution.

Table 1: Performance Measures: Control vs. Explanation Conditions (H1)

Measure	Control	Explanations	t	p	d
Game Score	125±18	114±23	-1.53	.136	-0.51
Learning Rate	5.4±8.8	9.2±7.5	-1.44	.158	0.50

Note: Explanations = Combined explanation conditions (Text n=14, Audio n=10). Learning rate = linear slope across sessions. Values shown as M±SD.

4 RESULTS AND DISCUSSION

We evaluated explanation effects on human-agent collaboration across three hypotheses: whether explanations improve performance (H1), whether modality matters (H2), and whether explanations improve subjective experience (H3). Our analysis reveals a nuanced picture: while explanations provided no immediate performance benefits for our gaming-expert participants, they influenced adaptation patterns and subjective perceptions in ways that illuminate both the promise and limitations of status-based XAI support.

4.1 Performance Effects: Expertise and Adaptation

An independent samples t-test comparing the control condition against combined explanation conditions showed a trend towards better performance of participants who did not receive any explanations (t(36)=-1.53, p=0.136, Cohen's d=-0.51), though this difference did not reach statistical significance. As shown in Table 1, the control condition scored higher (M=125.00, SD=18.19, n=14) than explanation conditions (M=114.17, SD=22.59, n=24), with a medium effect size suggesting practical relevance despite not reaching statistical significance.

The identified trend is directionally consistent with with Paleja et al.'s finding that expert performance can degrade with XAI support [10], as experienced users may quickly develop efficient mental models that explanations may disrupt. Our experiment population, made up of participants predominantly experienced with videogames (M=8.1 familiarity on 1-10 scale, SD=2.1), may have identified the optimal Counter-Circuit strategy – using the middle counter for rapid object transfers – rendering status-based explanations redundant once coordination patterns stabilized. Critically, we found no significant cognitive load differences across conditions (see Table 2), suggesting the performance trend stems from explanation interference with fluid coordination rather than raw processing demands.

However, examining *adaptation patterns* reveals a more compelling story. Analysis of learning rates (linear slopes across four sessions) revealed a clear monotonic pattern suggesting explanations accelerate adaptation: Audio participants improved fastest (M=11.20 points/session, SD=6.94), followed by Text (M=8.00, SD=7.85), then Control (M=5.39, SD=8.77). These results represented a 2.1× faster learning rate for Audio versus Control. While our one-way ANOVA

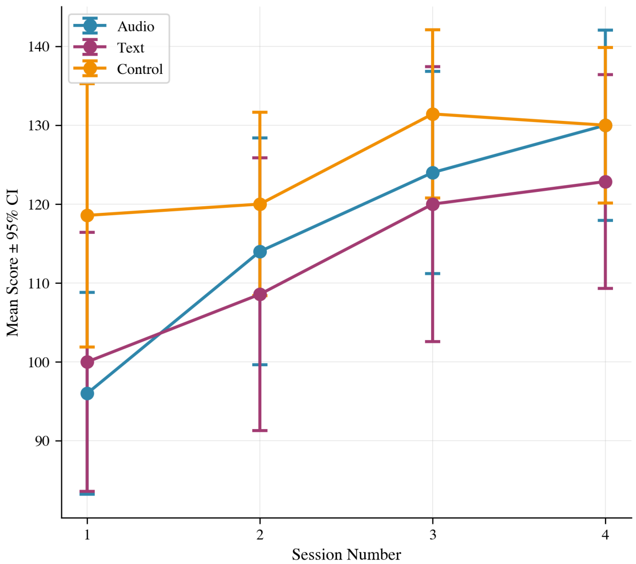

Figure 5: Learning curves by condition showing equivalent peak performance but differential adaptation rates. Audio explanations produced the steepest learning trajectory despite no final performance advantage.

did not reach statistical significance (F(2,34)=1.507, p=0.236), the Control versus combined explanations comparison approached significance with a medium effect size (t=-1.441, p=0.158, d=0.496). The Text versus Audio comparison also showed a small-to-medium effect (t=-1.032, p=0.313, d=0.427), demonstrating systematic differences in adaptation speed despite equivalent peak performance. Figure 5 illustrates these trajectories, with all conditions showing robust learning effects (r=0.335, p<0.001) but differing rates of improvement.

This pattern of equivalent peak performance alongside differential learning rates may suggest that explanations enhance human adaptation to agent behavior patterns rather than immediate task execution, though further research is needed to confirm this relationship. Expert teams may reach similar performance ceilings regardless of transparency, but explanations accelerate the mental model development that enables efficient coordination. The audio modality's showing of faster learning with equal peak scores to the Control condition supports our hypothesis that audio delivery might be promising for improved explanation delivery during fast-paced collaborative tasks.

4.2 Modality Effects: Salience vs. Intrusiveness

Comparing text versus audio explanations directly (H2) revealed equivalent performance (t(22)=0.329, p=0.745, d=0.136), with audio participants scoring marginally higher (M=116.00, SD=17.61) than text participants (M=112.86, SD=26.14). This negligible difference masks important salience effects: multiple text-condition participants reported not noticing explanations until later sessions, while audio explanations were universally perceived.

Table 2: Cognitive Workload: Text vs. Audio Explanations (H2)

NASA-TLX	Text	Audio	t	p	d
Mental Demand	3.1±1.4	3.2±1.3	0.05	.965	0.02
Effort	3.7±1.1	3.2±1.4	-0.95	.351	-0.40

Note: No significant differences after Bonferroni correction (α=0.0083). Values on 1-7 scale, M±SD.

> *Participant 12 (Text): "I didn't notice the agent's explanations. I only realized the agent was communicating in the last session, and that was the session where I managed to score the most points. Since I was very visually focused on the middle of the screen, I ended up not seeing the messages below."*

This feedback suggests text explanations could be easily disregarded, being treated as optional visual elements during fast-paced gameplay, while audio explanations demanded attention. This attentional difference may explain the modality's differential effects on learning rates and subjective measures (discussed below), even when both delivered identical content.

NASA-TLX analysis with Bonferroni correction (α = 0.0083) revealed no significant workload differences between modalities across six dimensions, as shown in Table 2. Mental demand was nearly identical (Text: M=3.12, SD=1.36; Audio: M=3.15, SD=1.34; t(22)=0.045, p=0.965, d=0.018). The largest observed difference was in effort ratings (Text: M=3.71, SD=1.13; Audio: M=3.23, SD=1.38; t(22)=-0.954, p=0.351, d=-0.395), which approached a small effect but remained non-significant after correction.

These findings contradict concerns that audio explanations would impose higher cognitive costs; instead, their greater salience may paradoxically reduce processing effort by making agent intentions immediately available without requiring visual attention shifts.

4.3 Subjective Experience: A Transparency Trade-off

One-way ANOVAs examining subjective measures (H3) across 13 Godspeed and Human-Agent Fluency subscales revealed no statistically significant differences after Bonferroni correction (α = 0.0038). However, two subscales showed theoretically important trends. Anthropomorphism ratings approached significance (F(2, 35) = 2.703, p = 0.081, η^2 = 0.134), with audio participants rating the agent as less anthropomorphic (M=2.44, SD=0.65) compared to text (M=3.11, SD=0.76) and control (M=3.16, SD=0.96) conditions. Trust ratings showed a parallel pattern (F(2, 35) = 2.464, p = 0.100, η^2 = 0.123), with audio participants reporting lower trust (M=2.20, SD=0.63) versus text (M=2.89, SD=1.00) and control (M=3.11, SD=1.21).

This transparency trade-off – where explanations trended toward making the agent seem less human-like (though not reaching statistical significance, p=0.081) – contrasts with Paleja et al.'s finding that explanations consistently improved subjective ratings [10]. While our results remain preliminary, they may point to a fundamental limitation of status-based explanations from model-free policies:

the agent frequently changed subtask selections mid-execution, creating mismatches between stated intentions and observed actions. Participants noticed this inconsistency:

> *Participant 2 (Text): "In the last two sessions, the agent kept saying it was going to do A and then did B or C, something I didn't see in the first two sessions. This caught my attention and made me pay more attention to what it was actually going to do, instead of what it said it was going to do."*

By making the agent's decision-making process visible, explanations also exposed its artificial nature. The HA^2 Manager policy runs every timestep, responding reactively to observations without maintaining commitments to previously selected subtasks, which is a kind of behavior natural for model-free reinforcement learning but confusing when communicated as human-like intentions. Rather than building trust through transparency, explanations highlighted the gap between human coordinative communication (predictive and committal) and reactive ML policy outputs.

4.4 The Limits of Status-Based Explanations

Participant feedback reveals two critical gaps in our status-based approach. First, *predictive value*: participants wanted explanations that enabled proactive coordination, not just status reports:

> *Participant 27 (Audio): "The agent performed tasks in order to complete the team's goal while verbalizing its next action, so there is cooperation. However, it wasn't possible for me to actually coordinate my actions with the agent, since it could only guess what I wanted to do and I couldn't tell it to act differently. Ideal cooperation would be to divide tasks in a way that accelerates the process."*

> *Participant 16 (Audio): "I think the human-agent interaction would be better if the agent were capable to notice a mistake by the human part and wait or suggest a fix action to the player. For example, the agent could say 'Get out the way' instead of 'I'm trying to serve a plate.'"*

Second, *coordinative utility*: expert human teams in Overcooked-AI develop sophisticated patterns involving task division negotiation, error recovery, and dynamic reallocation—mechanisms our status explanations couldn't support. Human-human communication in this context is inherently bidirectional and strategic; our unidirectional status reports provided information but not coordination affordances.

These limitations stem from the model-free nature of hierarchical RL policies. Without explicit world models or planning capabilities, HA^2 cannot generate predictive explanations ("I will need to do X next") or intentional explanations ("I am doing X to achieve Y"). This represents a fundamental constraint: intrinsically explainable architectures provide authentic insights into current decision-making but cannot support the richer communication that characterizes human collaboration.

It is important to note that while these limitations provide a theoretically coherent account of our observed trends, the non-significant statistical results prevent us from definitively attributing the performance and perception patterns to these model-free constraints. Nevertheless, participant feedback clearly identifies gaps in status-based explanations regardless of their measurable impact, suggesting these limitations warrant attention in future XAI system design.

4.5 Implications and Future Directions

Before discussing implications, we must acknowledge the exploratory nature of this work. While our study was designed to detect large effects based on pilot testing and similar prior research [10], several of our key findings showed meaningful effect sizes without reaching statistical significance: performance differences (d=-0.51, p=0.136), learning rate improvements (d=0.50, p=0.158), and anthropomorphism reductions (η^2=0.134, p=0.081). These patterns suggest potentially important relationships that require replication in adequately-powered studies before drawing firm conclusions. We therefore frame the following as preliminary insights emerging from this first systematic investigation of intrinsically explainable ML policies in an established HAT benchmark, rather than definitive findings.

With this caveat, three key insights emerge from this exploratory investigation. First, explanation utility depends critically on the match between user expertise, task complexity, and explanation sophistication. Gaming experts coordinating in Counter-Circuit showed trends towards better performance without explanations, suggesting simple collaborative tasks may not require XAI support once teams develop efficient patterns. Future work should evaluate hierarchical explanations in more complex environments, using more layouts of Overcooked-AI or with populations different from that of our study (such as gaming novices and/or older adults).

Second, audio modalities show promise for ensuring explanation salience in visually demanding tasks, despite providing no immediate performance advantage. The faster learning rates under audio delivery suggest that modality choices matter for adaptation even when they don't affect peak performance. Refining audio delivery, through the use of more natural voices instead of basic TTS, could enhance this effect. While our sample size (n=38) provided adequate power to detect large effects, several trends approached significance with medium effect sizes (learning rate improvements: d=0.50, p=0.158; modality differences: d=0.43, p=0.313). Larger-scale studies could clarify whether these patterns represent meaningful but subtle effects or noise in the data.

Third, status-based explanations from model-free policies may be insufficient for supporting sophisticated coordination, based on both the observed trends and participant feedback. Model-based architectures that can generate stable, predictive explanations may better align with human expectations. Alternatively, bidirectional communication systems that allow humans to provide directives or negotiate task division could address the coordinative limitations we observed.

A methodological consideration for future research is the impact of our between-subjects design. While this approach enabled assessment of explanation effects without carryover or practice effects across conditions, between-subjects designs might have lower statistical power for detecting differences compared to within-subjects designs. This may have particularly affected our subjective measures (e.g., anthropomorphism, trust, NASA-TLX workload), where within-person comparisons could reveal modality preferences more sensitively. Future work employing within-subjects designs where participants experience multiple conditions could provide stronger tests of modality effects on subjective perceptions, though such designs would need to carefully control for order effects and learning.

Despite the statistical limitations, our work demonstrates that intrinsically explainable machine learning policies can generate authentic explanations in benchmark collaborative environments without sacrificing performance, establishing a methodological foundation even as the specific effects observed require further investigation. While simple status explanations showed limited utility in our context, this methodological contribution establishes a foundation for more sophisticated XAI systems that leverage hierarchical structures for richer collaborative support.

5 CONCLUSION

This work establishes both the potential and constraints of leveraging intrinsically explainable machine learning architectures for collaborative AI systems. We provide the first systematic evaluation of XAI support generated from state-of-the-art learned policies in an established human-agent teaming benchmark, demonstrating that hierarchical reinforcement learning architectures can serve as both high-performing agents and sources of authentic explanations without sacrificing either capability. While several key findings did not reach statistical significance due to our exploratory sample size (n=38), the observed patterns provide preliminary insights that can guide future adequately-powered research.

Our findings reveal a nuanced picture of explanation utility among gaming experts. Simple status explanations from the HA^2 hierarchical policy showed trends toward lower immediate performance – the control condition trended toward better performance (d=-0.513, p=0.136) – yet audio explanations showed trends toward faster learning rates (d=0.496, p=0.158). This pattern may suggest that XAI support enhances human adaptation to agent behavior rather than immediate task execution, with transparency possibly accelerating human mental model development of the agent's behavior. The audio modality's advantage of greater salience without increased cognitive load indicates that modality design matters for ensuring explanations reach users in visually demanding collaborative tasks.

However, our work also reveals potential limitations of status-based explanations from model-free policies. While not reaching statistical significance (p=0.081, p=0.100), trends suggested participants rated agents with explanations as less anthropomorphic and trustworthy, contrasting with prior XAI findings [10]. By making decision-making visible, explanations also exposed the agent's artificial nature: frequent mid-execution subtask changes that are natural for reactive RL policies but confusing when communicated as human-like intentions. This transparency trade-off highlights a fundamental challenge: explanation systems must account for how policies actually execute, not just what they currently intend.

Our methodological contributions enable future XAI research in collaborative contexts. We demonstrate that: (1) intrinsically explainable architectures can generate authentic real-time explanations in benchmark environments; (2) audio delivery may ensure greater salience in fast-paced tasks; (3) trigger-based systems can manage communication frequency while maintaining naturalness; and (4) model-free policies may impose fundamental constraints on explanation richness that warrant further investigation. These insights suggest that effective collaborative XAI requires matching explanation sophistication to both user expertise and task complexity.

Future work should evaluate hierarchical explanations with novice users, in multiple environments, and using model-based architectures that can generate predictive rather than purely reactive explanations. Longitudinal studies with more sessions could clarify whether learning rate advantages eventually translate to sustained performance benefits. Most critically, bidirectional communication systems that enable humans to provide directives or negotiate task division may address the coordinative limitations we observed in unidirectional status reports.

Our work provides empirical evidence that hierarchical RL architectures offer a promising foundation for intrinsically explainable collaborative AI. While simple status explanations showed limited utility in our context, this methodological contribution establishes a foundation for systematically evaluating XAI approaches while maintaining the performance standards expected in modern human-agent teaming applications.

Limitations: Our convenience sampling yielded a homogeneous sample of predominantly young, gaming-experienced participants from the academic community. While this enabled baseline findings suggesting XAI support does not universally improve collaboration among experienced users, generalizability to novice users, older adults, or populations with less gaming experience remains unknown. The sample's gaming expertise may have contributed to the observed trend that explanations did not improve immediate performance, as expert users may develop efficient coordination strategies that render simple status explanations redundant or even intrusive. Additionally, our sample size (n=38), while adequate for detecting large effects, was underpowered for the medium effects we observed. This statistical limitation necessitates treating our findings as preliminary patterns requiring replication in larger, adequately-powered studies before drawing firm conclusions about XAI effectiveness in human-agent collaboration.

SUPPLEMENTARY MATERIALS

Supplementary materials include: (1) complete trigger-based explanation templates with variable mappings and implementation details (Appendix A) and (2) full survey instruments for all measures used in the study (Appendix B).

ACKNOWLEDGMENTS

This study was approved by PUC-Rio's Research Ethics Committee (Protocol SGOC 545928, approved June 11, 2025). This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001.

This paper was accepted as an Extended Abstract at AAMAS 2026.

REFERENCES

[1] Stéphane Aroca-Ouellette, Miguel Aroca-Ouellette, Katharina von der Wense, and Alessandro Roncone. 2025. Implicitly Aligning Humans and Autonomous Agents through Shared Task Abstractions. arXiv:2505.04579 [cs.MA] https://arxiv.org/abs/2505.04579

[2] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International journal of social robotics* 1, 1 (2009), 71–81.

[3] Micah Carroll, Rohin Shah, Mark K. Ho, Thomas L. Griffiths, Sanjit A. Seshia, Pieter Abbeel, and Anca Dragan. 2020. On the Utility of Learning about Humans for Human-AI Coordination. https://doi.org/10.48550/arXiv.1910.05789 arXiv:1910.05789 [cs].

[4] Maaike Harbers, Jeffrey M. Bradshaw, Matthew Johnson, Paul Feltovich, Karel Van Den Bosch, and John-Jules Meyer. 2011. Explanation and Coordination in Human-Agent Teams: A Study in the BW4T Testbed. In *2011 IEEE/WIC/ACM International Conferences on Web Intelligence and Intelligent Agent Technology*. IEEE, Lyon, France, 17–20. https://doi.org/10.1109/WI-IAT.2011.83

[5] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, 139–183.

[6] Guy Hoffman. 2019. Evaluating fluency in human–robot collaboration. *IEEE Transactions on Human-Machine Systems* 49, 3 (2019), 209–218.

[7] Yi Loo, Chen Gong, and Malika Meghjani. 2023. A Hierarchical Approach to Population Training for Human-AI Collaboration. https://doi.org/10.48550/arXiv.2305.16708 arXiv:2305.16708 [cs].

[8] Drew McDermott. 1978. Tarskian semantics, or no notation without denotation! *Cognitive Science* 2, 3 (1978), 277–282.

[9] Joseph E Mercado, Michael A Rupp, Jessie YC Chen, Michael J Barnes, Daniel Barber, and Katelyn Procci. 2016. Intelligent agent transparency in human–agent teaming for Multi-UxV management. *Human factors* 58, 3 (2016), 401–415.

[10] Rohan Paleja, Muyleng Ghuy, Nadun Ranawaka Arachchige, Reed Jensen, and Matthew Gombolay. 2021. The utility of explainable ai in ad hoc human-machine teaming. *Advances in neural information processing systems* 34 (2021), 610–623.

[11] Fatai Sado, Chu Kiong Loo, Wei Shiung Liew, Matthias Kerzel, and Stefan Wermter. 2023. Explainable goal-driven agents and robots-a comprehensive review. *Comput. Surveys* 55, 10 (2023), 1–41.

[12] Natalie Sebanz, Harold Bekkering, and Günther Knoblich. 2006. Joint action: bodies and minds moving together. *Trends in cognitive sciences* 10, 2 (2006), 70–76.

[13] D. J. Strouse, Kevin R. McKee, Matt Botvinick, Edward Hughes, and Richard Everett. 2022. Collaborating with Humans without Human Data. https://doi.org/10.48550/arXiv.2110.08176 arXiv:2110.08176 [cs].

[14] Joshua B Tenenbaum, Charles Kemp, Thomas L Griffiths, and Noah D Goodman. 2011. How to grow a mind: Statistics, structure, and abstraction. *science* 331, 6022 (2011), 1279–1285.

[15] Chenxu Wang, Zilong Chen, and Huaping Liu. 2024. On the Utility of External Agent Intention Predictor for Human-AI Coordination. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*. 2546–2548.

[16] Rui Zhang, Wen Duan, Christopher Flathmann, Nathan McNeese, Guo Freeman, and Alyssa Williams. 2023. Investigating AI Teammate Communication Strategies and Their Impact in Human-AI Teams for Effective Teamwork. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (Sept. 2023), 1–31. https://doi.org/10.1145/3610072